

# Project Proposal

**Project Title:** *"Design and Implementation of an Intelligent Automatic Dustbin using Ultrasonic Sensors and Servo Based Control"*

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## Project Overview

This project involves the design and development of an intelligent automatic dustbin system that enhances hygiene and waste management through sensor-based automation. The system uses ultrasonic sensors to detect user presence and monitor the fill level of the dustbin, enabling automatic lid operation through a servo motor. Integrated visual feedback using LEDs provides real-time status indication, ensuring safe, efficient, and contactless disposal of waste. The proposed solution demonstrates the practical application of sensors, actuators, and feedback control in a real-world embedded system.

## Motivation and Significance

Manual operation of public and household dustbins often leads to hygiene issues, inconvenience, and the spread of germs due to frequent physical contact. Existing automatic dustbins either lack fill-level awareness or continue operating even when full, resulting in overflow and system misuse. This project addresses these limitations by introducing an intelligent, sensor-based control mechanism that ensures hygienic operation, prevents overfilling, and improves user safety through real-time feedback. **Scope and Applications:** The scope of this project includes the design of a compact, low-cost automated dustbin capable of user detection, fill-level monitoring, and safe lid control. The system is suitable for deployment in households, educational institutions, offices, hospitals, and public places. Additionally, it serves as an effective educational platform for demonstrating sensors, actuators, and feedback systems in embedded applications.

## Working Principle

The system operates based on real-time feedback obtained from two ultrasonic sensors interfaced with a microcontroller. The first ultrasonic sensor detects the presence of a user or object near the dustbin and triggers the servo motor to open the lid automatically. The lid remains open for a predefined delay period to allow waste disposal.

A second ultrasonic sensor mounted inside the dustbin continuously monitors the fill level. When the bin reaches its maximum capacity, the system disables the lid-opening mechanism and activates a red indicator LED to prevent further use. Before closing the lid, the system rechecks for object presence to ensure safety, while a yellow LED provides a visual warning during the closing delay. A green LED indicates normal operation when sufficient capacity remains.

## System Components

- **Microcontroller:** Arduino Nano – central processing and control unit

- **Ultrasonic Sensors:**

- Sensor 1 – Detects user/object near the dustbin (external detection)
- Sensor 2 – Monitors internal fill level (feedback sensor)

- **Servo Motor(Metal Gear MG90s):** Controls lid opening and closing, operates based on sensor feedback

- **LED Indicators:**

- Red LED – Bin full, prevents lid operation
- Yellow LED – Warning: Lid will close soon
- Green LED – Bin has capacity, normal operation

- **Power Supply:** Provides stable voltage to sensors, actuators, and microcontroller

- **Structural Enclosure:** Dustbin body with lid mechanism

**Feedback Components:** Sensors and LEDs provide feedback to the microcontroller to control servo operation and inform users about the bin status.

## Conclusion

The proposed intelligent automatic dustbin demonstrates a practical application of sensors, actuators, and feedback control in a real-world embedded system. By integrating ultrasonic sensors for user detection and fill-level monitoring with servo-based lid control and LED indicators, the system ensures hygienic, efficient, and safe waste disposal. This project highlights the effectiveness of automated feedback mechanisms in enhancing usability and preventing overfilling, making it a robust and scalable solution for households and public spaces.

## Future Enhancements

- **Wireless or IoT integration** for remote monitoring and notifications.
- **Advanced fill-level detection** using weight sensors or capacitive sensing for higher accuracy.
- **Integration of AI-based predictive algorithms** for usage patterns.
- **Additional safety features** such as obstruction detection during lid closure.
- **Expansion to smart waste management systems** in smart cities or offices.